Pilot user guide

Pilot is a ZX Spectrum tape loader for Resolume Arena and Avenue, as an FFGL effect, and for DaVinci Resolve, Vegas, Nuke and Natron as an OpenFX effect and transition (see OpenFX). Your clip arrives the way a Spectrum loaded a screen: in screen-memory order, at the baud rate, monochrome first and colour last, with the border striped by the loading signal itself.

In practice it's a transition, but it isn't a shape moving across the frame. It's an **address order**, and the pattern it makes is the one the hardware made rather than one anybody chose. Anyone who ever waited for a Spectrum game to load will know it at once.



Rendered by the plugin's own offline harness rather than captured from Resolume.

> **Before you rely on this:** it's at **v0.2.0**, which adds the OpenFX build for Resolve, Vegas, Nuke and Natron, and still early days. An offline harness driving the real plugin class proves the address order agrees with an independent derivation for all 6144 addresses, bitwise, and that the rendered frame matches it for all 49,152 screen pixels at 1920×1080 and 640×480, at six points in the load, with zero disagreements; all 14 controls are confirmed to change the picture.
>
> It has **never been loaded into Resolume on macOS**, so how the controls *look* in the inspector is untested. On Windows, a build of v0.1.0 loads, registers and renders in Resolume Arena 7.27.1, with every control matching what the plugin declares. That was on software rendering, so it says nothing about a GPU.
>
> The OpenFX build has been in one real host, DaVinci Resolve Studio 21.1 on macOS. As a Fusion tool it renders exactly what the test host renders, and as a transition on the Edit page it plays in the right order, fading in from the outgoing clip and out to the incoming one, though its first and last frames come close to the clips rather than matching them exactly (see OpenFX). It has **never been in Vegas, Nuke or Natron**, and the Windows and Linux OpenFX builds have never rendered in any host.
>
> **Try it on a spare layer first**, and please report anything that misbehaves.
>
> This codebase was created with AI assistance, directed and reviewed by a human author.

Installing

Drop the plugin into Resolume's effects folder and restart Resolume:

```
macOS    ~/Documents/Resolume Arena/Extra Effects/
Windows  %USERPROFILE%\Documents\Resolume Arena\Extra Effects\
```

Avenue uses the same layout under its own folder name. It then shows up in the effects browser as **SW Pilot**.

The macOS download is a universal build (Apple silicon and Intel) as a `.dmg` or a `.zip`; Windows is x64, as an installer or a `.zip`. The macOS build is Developer ID-signed and notarised, so it just loads. The Windows build isn't code-signed: the plugin file itself loads normally, and only the installer trips SmartScreen, once: **More info** → **Run anyway**.

Start here

Put SW Pilot on a layer with a clip on it and leave every control alone. You get a Spectrum screen **part way through loading**: a black-and-white picture of your clip filling in, in bands, inside a border

of yellow and blue stripes. That's the ZX 48 at its own rate, with **Progress** at 45% and **Sync** on Manual.

There's no trigger button. **Where the load has got to is Progress**, and nothing else:

- **Manual** (the default) — drag Progress from 0 to 1 and the tape plays through; drag it back to 0 and the load starts again. Scrub it by hand, map it to a fader, or animate it with Resolume's own parameter animation.
- **Clip time**, **Beat** and **Bar** — the load runs by itself and **loops**: when it reaches the end it wraps straight back to the start and loads again. Progress is still live, as an offset: it sets where in the load the clock starts from.

Watch one whole load and three things happen that nobody designed. The picture doesn't wipe down the screen: it arrives in three thirds, each filling in as an eight-line venetian blind. It stays **monochrome** until about 89% of the way through, and then the colour lands all at once, because the colour bytes sit after the picture bytes on the tape. And the border stripes are the tape signal itself, so they move with the data.

The Machine group

Type — which loader: **ZX 48**, **ZX 128**, **C64 turbo** or **Amstrad**. This sets the nominal baud rate and the border's colours and style (see The machines). The screen itself is a Spectrum display file whatever you pick.

Baud — a trim around the machine's own rate rather than an absolute number: ×¼ at the bottom, the machine's real rate at the **centre** (the default), ×4 at the top. It sets how fast the border stripes run and how long a load takes in Clip time sync. **It doesn't change how fast the picture arrives in Manual.** That's Progress's job, on purpose: a real ZX 48 load is 37 seconds, and a 37-second transition you can't scrub is no use to anybody.

Ink and **Paper** — the power-on attribute: the two colours the picture wears **before its own colours arrive**. Each is one of the Spectrum's eight colours: Black, Blue, Red, Magenta, Green, Cyan, Yellow, White. A Spectrum powers up black ink on white paper, and so does this. They're also the colours the loading-error message is printed in, and Paper is what the Paper background shows.

Bright — Off, **Auto** (the default) or On. BRIGHT was one bit per character cell on the real machine, lifting that cell's colours from 84% to full level. **Off** never sets it, so white is a light grey. **On** sets it on every cell, and on the power-on Ink and Paper too. **Auto** lets the picture decide, cell by cell: a cell whose brightest channel is above half way between the two hardware levels goes bright.

The Load group

Progress — where the load has got to, 0 to 1. The first stretch of it is the pilot tone (see **Pilot Length**), where the border is running and no picture has arrived yet; the rest is the tape, 6912 bytes of it. In the sync modes it becomes the offset the clock is added to.

Sync — what drives the load:

Sync	One whole load takes
Manual	As long as you take to move Progress from 0 to 1. Nothing moves by itself.
Clip time	As long as the real tape: 6912 bytes at the current baud rate, plus the pilot tone.
Beat	One beat of Resolume's tempo.
Bar	One bar, counted as four beats.

Clip time is where the faithful behaviour lives. The tape really does run at the baud rate: at the defaults a ZX 48 load is about 37 seconds of data, 40 with its pilot tone. The other machines at their own rates take 25–30 seconds, and Baud stretches any of them from about a quarter of that to four times it. It runs from the clock Resolume hands the plugin, and loops.

Beat and **Bar** follow Resolume's BPM and bar position, so a load starts on the beat or the bar line when Progress is at 0. With no tempo from the host they assume 120 BPM.

Error Rate — the chance that each block of the tape fails, from nothing (the default) up to one in two at the top of the slider. The tape is 27 blocks of 256 bytes. When a block fails, the Spectrum's own report is printed over whatever had arrived:

```
R Tape loading error, 0:1
```

The picture then **holds** for one block's worth of tape. Then the load starts again from an empty screen with whatever tape is left. The failures are seeded: the same settings fail in the same places every time, so scrubbing Progress back over a failure shows the same failure again, not a new roll of the dice. Even a low setting fails often over 27 blocks; halfway up the slider almost no load finishes.

Message On — whether the report is printed when a block fails. It does nothing at all unless Error Rate is up and a block has just failed. With it Off, a failure still holds and restarts the picture; it just does it silently.

The Border group

Border On — whether there's a border at all. Off isn't a colour: it takes the border away, and the screen fills the whole composition. **Border Width** then has nothing to act on.

Border Width — how much of the composition the border takes off **each edge**, from none up to a quarter. The default is about 8%, which is close to what a Spectrum's border was. The clip is squeezed into whatever is left, aspect and all, so the whole of it arrives framed, and on a 16:9 composition a Spectrum pixel is wider than it is tall, as it was on a widescreen television.

Pilot Length — how much of the Progress range is spent on the pilot tone before the first byte lands, from none to half of it. The default is about 8%. During the pilot the screen shows only the **Background** and the border shows the machine's pilot colours (red and cyan on a ZX 48), changing to its data colours the moment the first byte arrives. At zero there's no pilot and the picture starts at once.

The machines

The border isn't decoration. The Spectrum's border was told the state of the tape signal and the beam painted it as it scanned, so the stripes are horizontal and there are **baud ÷ 50** pairs of them from the top of the frame to the bottom: 30 on a ZX 48, 45 on a ZX 128, 40 on the Amstrad. Turn **Baud** up and there are more, thinner stripes.

Type	Rate	Pilot tone	Data
ZX 48	1500 baud	red / cyan stripes	yellow / blue stripes
ZX 128	2250 baud	magenta / green stripes	white / black stripes
C64 turbo	2400 baud	the whole border flashing	the whole border flashing
Amstrad	2000 baud	bright cyan / blue stripes	bright red / yellow stripes

Only the ZX 48 models a documented routine: its rate and its red/cyan then yellow/blue are the 48K ROM loader's own. The other three are just *looks* in the same shape. The ZX 128 is what 128K games actually shipped with (a fast loader on Spectrum hardware) rather than the 128's ROM, which loaded exactly like the 48. The C64 turbo's border is one colour at a time, changing once per byte at random and never to black, so it reads as a flicker rather than stripes. I chose the Amstrad's colours to be distinct rather than measuring them from a machine.

The loading error

A real Spectrum didn't clear the screen when a load failed. It printed the report over whatever had arrived and stopped. Pilot does the same: while the message is up, the picture freezes at what the failed attempt had reached; when it goes, the retry takes over from an empty screen. The report is drawn along the bottom character row of the screen, in the power-on **Ink** on **Paper**.

Because the retry starts with only the tape that's left, a load with **Error Rate** up can reach Progress 1 part-loaded. On a real machine you'd rewind the tape, sigh, and press play again. In Manual that's dragging Progress back; in the sync modes it happens on the next loop.

Output

Background — what an address that **hasn't arrived yet** shows:

- **Paper** (the default) — the power-on Paper colour, which is what a real machine showed.
- **Black** — black.
- **Clip** — the live clip, untouched. This turns the whole thing into a colour-and-quantise transition: the Spectrum version of the picture loads over the real one, in the Spectrum's order.

It changes the screen area only, never the border.

Mix — dry/wet against the untouched clip. At **0** the output is the input, byte for byte, whatever everything else is set to (measured, not assumed). At full Mix the output is opaque.

How it works

The Spectrum's screen is 6144 bytes of picture followed by 768 bytes of colour, and the picture isn't stored top to bottom. The byte holding a pixel lives at

```
third × 2048  +  line × 256  +  character row × 32  +  column byte
```

That's which third of the screen, then which of the eight lines inside a character cell, then which character row within that third. Fill the memory from the start of the tape to the end and the screen fills in that order: the first line of every character row in the top third, then the second line of each, and so on, then the middle third, then the bottom. That's the venetian blind. Pilot does nothing but fill the addresses in order.

The colour comes last because the 768 attribute bytes come last: one byte per 8×8 cell, giving that cell one ink, one paper and a BRIGHT bit. To make those, Pilot shrinks the clip to the Spectrum's 256×192, splits each cell into its darker and lighter halves, and gives each half the nearest of the eight colours. Which half a pixel belongs to is settled from the picture, so the pattern of the monochrome phase doesn't change when colour arrives; only its two colours do.

Performance

Cheap, and nearly independent of the composition size, because the two expensive passes always run at 256×192 and 32×24. Measured on an M4 Max:

Composition	ms/frame	Share of a 60 fps frame
1280×720	0.09	0.5%
1920×1080	0.21	1.2%
3840×2160	0.36	2.2%

Nothing has been timed on Windows or on an Intel Mac.

The OpenFX build renders on the CPU and costs more: about 5 ms a frame at 1920×1080 with 16 threads on the same machine (1.8 ms at 1280×720, 11 ms at 3840×2160), and about seven times that on a single thread.

Known limits

- **Never loaded into Resolume on macOS.** Whether the four groups read sensibly in the inspector, whether Ink and Paper show as colour lists, and how Arena's own textures behave are all untested.
- **Baud doesn't drive the reveal in Manual.** Use Clip time if you want the tape to take as long as a real one.
- **The four machines are four looks.** Only the ZX 48 models a real routine, and none has been compared against a real tape or an emulator. Every machine loads a Spectrum screen.
- **Colours are recomputed every frame.** On moving footage a cell's two colours change *during* the load, which a real tape never could, since its screen was converted once, before it was recorded.
- **Clip time runs off Resolume's clock** and loops, rather than following the clip's own playhead.
- **Bar is always four beats.**
- **At the end of an error hold the picture jumps**, for one frame, to the retry's progress. A real machine drew the retry over the old picture.
- **No factory presets.**
- **The OpenFX build has been in one real host only**: DaVinci Resolve Studio 21.1 on macOS, as a Fusion tool and as a transition on the Edit page. It has never been in Vegas, Nuke or Natron, nor tried as an effect on Resolve's Edit or Color page, and the Windows and Linux builds have never rendered in a host. In Resolve a Fade transition's first and last frames are close to the clips, not exactly them (see OpenFX). It has no Beat or Bar sync.

- **There's a browser demo** at pilot-demo.stoatworks-labs.com. It's a port to a web page rather than the plugin: the shaders run in WebGL2 and any CPU half is rewritten in JavaScript. The page lists what it doesn't reproduce.

OpenFX — Resolve, Vegas, Nuke, Natron

The same effect builds as an OpenFX plugin for DaVinci Resolve (Edit and Color pages, and Fusion), Vegas Pro, Nuke and Natron. It's a separate download (`pilot-ofx-macos-universal.zip`, `pilot-ofx-windows-x86_64.zip` or `pilot-ofx-linux-x86_64.zip`, from v0.2.0 on), and it goes in the OpenFX folder rather than Resolume's. Unzip it and copy `Pilot.ofx.bundle` into:

```
macOS    /Library/OFX/Plugins/
Windows  C:\Program Files\Common Files\OFX\Plugins\
Linux    /usr/OFX/Plugins/
```

Restart the host and it shows up as **Pilot**, under **Stoatworks**, in two places:

- **As an effect** on a clip, with the controls this guide describes. Progress is keyframeable; **Sync** offers **Manual** and **Clip time**, where the tape runs at the baud rate along the timeline and loops.
- **As a transition** between two clips. The transition's own position drives the tape: the incoming clip is the picture that loads, and the outgoing clip is what every address that hasn't arrived yet shows. **Background** starts on **Clip** here (choose Paper for a machine's blank screen instead), and **Mix** fades the load against the outgoing clip. There's no Progress or Sync, because the transition is the clock.

The transition has one more group, **Ends**, which decides how it starts and finishes:

- **Fade** (the default) — the transition starts on exactly the outgoing clip and finishes on exactly the incoming one, so it sits on a timeline without a jump at either edit. Over the first **End Length** it fades from the outgoing clip into the empty Spectrum screen and its striped border; the tape loads in the middle; and over the last End Length it fades from the loaded screen into the incoming clip. End Length is 0.15 by default (the first and last 15%) and goes up to 0.5, where the load happens all at once at the midpoint.

  "Exactly" depends on the host asking for the very start and end of the transition, and **DaVinci Resolve doesn't**: it asks for the middle of each frame. So in Resolve the first frame is already a little way into the fade (a faint border over the outgoing clip), and the last isn't quite the incoming clip; the edit then cuts to the clean clip. In a one-second transition at 24 fps, with End Length at its default, each end is about 5% of the way into its fade, so the last frame is about 95% the incoming clip. A longer transition or a longer End Length makes that step smaller.

- **Cut** — the raw load, from the first frame to the last, as Resolume shows it: the border is there from the very first frame, and the last frame is the fully loaded Spectrum screen, after which

the edit cuts to the real incoming picture.

What the OpenFX build doesn't have:

- **Beat and Bar sync.** An OpenFX host doesn't give a plugin a tempo, so those two modes aren't offered.

Resolve's Fusion page reports the frame rate on the effect but not on its clips; the plugin reads the effect's, and assumes 24 fps only where a host reports none. So in Fusion, as on the Edit and Color pages, Clip time and the border's stripes follow the timeline's own rate.

It renders on the CPU rather than the GPU (about 5 ms a frame at 1920×1080 on a 16-thread Apple M4 Max, about 35 ms on a single thread), and the picture matches the Resolume build's except for the odd pixel sitting exactly on the edge between two Spectrum pixels or two border stripes, or exactly on its cell's brightness threshold, which can take the neighbouring colour: typically a few hundred pixels in a 1920×1080 frame.

If it looks like it is doing nothing

- **Mix** at zero is the clip, exactly as it came in.
- **Progress** inside the pilot tone shows only the Background and the border. Push it past **Pilot Length**.
- **Background** on **Clip** with Progress near 0 looks like the clip plus a border, because almost nothing has arrived yet.
- **Message On** does nothing until **Error Rate** is up and a block fails.

A shader that won't compile looks just like a plugin that does nothing, with the actual message in the log:

```
macOS    ~/Library/Logs/pilot/pilot.YYYY-MM-DD.log
Windows  %LOCALAPPDATA%\pilot\logs\pilot.YYYY-MM-DD.log
```

About

The **About** group at the bottom shows the plugin's name and version, with a row of buttons that open its links (the project page, the source and so on) in your browser.

Reporting something

github.com/stoatworks-labs/pilot/issues. A screenshot, the composition's resolution and the settings you were on is usually enough.